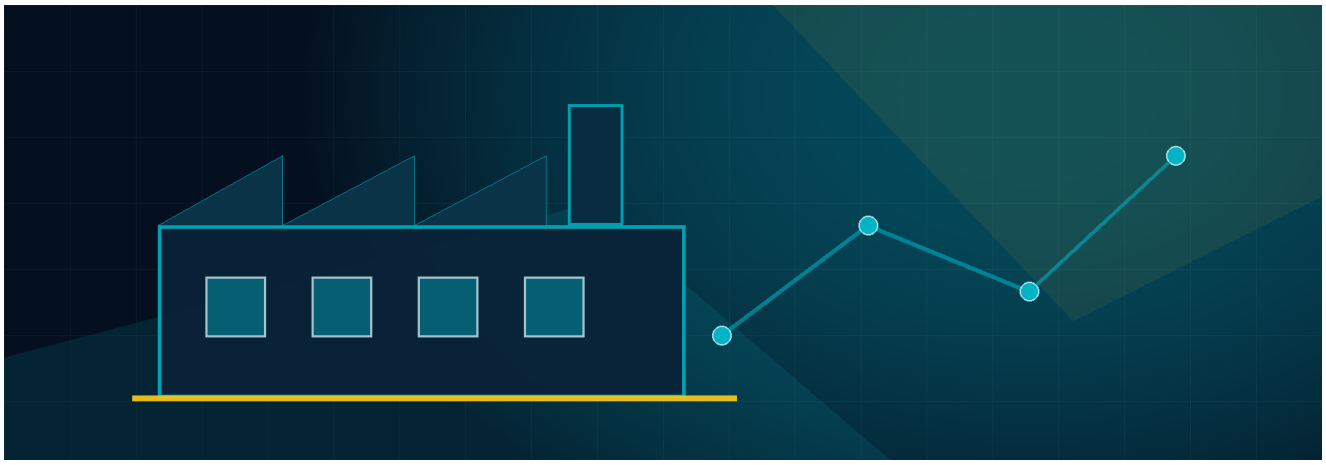
ROALLA



EXECUTIVE INDUSTRIAL INVESTMENT GUIDE

Industrial Technology Vendor Selection and Business Case Guide 2026

A decision framework for selecting technology partners and protecting lifecycle investment value

Prepared for Canadian owners, boards, chief executives, operations, engineering, supply chain, finance, procurement, technology, security, legal and transformation leaders

Prepared by ROALLA Transformation and Technology Advisory

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Purpose

This guide helps leadership evaluate industrial software, automation, equipment, integration and service partners using one evidence standard. It connects operational requirements, architecture, safety, cybersecurity, implementation, support, commercial terms, lifecycle cost and benefits so vendor selection is an investment decision rather than a feature comparison.

EXECUTIVE DECISION DASHBOARD

Management View at a Glance

The preferred vendor is not necessarily the lowest bid or strongest demonstration. Management should select the option that can meet verified operating requirements, integrate safely, survive lifecycle change, be supported locally and produce risk-adjusted value under enforceable commercial terms.

Executive question	ROALLA view	Decision implication
Are requirements decision-ready?	**PREPARE**	Translate operating outcomes, volumes, exceptions and constraints into weighted, testable requirements.
Will demonstrations prove fit?	**DECIDE**	Use client data and scripted scenarios rather than vendor-selected feature tours.
Can the vendor support the lifecycle?	**CONDITIONAL**	Validate team, references, service, parts, product roadmap, finances and partner dependency.
Do contract terms protect value?	**PREPARE**	Tie milestones, acceptance, payment, service, data, change and remedies to outcomes.
What should management authorize?	**PURSUE**	A vendor-neutral requirements, selection and business-case process.

Recommended Management Posture

Proceed through a small number of explicit decision gates. Preserve evidence as work occurs, separate discovery from implementation, and require each investment to demonstrate value, feasibility, control effectiveness and accountable ownership before scale.

EXECUTIVE DECISION MODEL

The Eight Industrial Vendor Decision Gates

A defensible selection begins with the operating problem and acceptance evidence. Requirements, demonstrations, references, architecture, contracts and the business case must test the same outcomes and risks.

Stage	Gate	Required conclusion	Decision evidence
01	Outcome	The investment decision is explicit	Operating baseline, target, scope, constraints and sponsor
02	Requirements	Needs are prioritized and testable	Must, should, option, volumes, exceptions and acceptance
03	Architecture	The solution fits the environment	Systems, data, interfaces, infrastructure, identity and lifecycle
04	Operational fit	Real workflows can be supported	Normal, peak, degraded, exception, safety and maintenance scenarios
05	Vendor capability	Delivery and support claims are evidenced	Team, references, methods, capacity, partners and financial viability
06	Risk	Security, safety and dependency are controlled	Threats, access, data rights, compliance, continuity and exit
07	Commercial	Terms align incentives and accountability	Scope, price, change, acceptance, warranty, service and remedies
08	Economics	Lifecycle value is credible	Total cost, ramp, benefits, sensitivity, contingency and residual value

Decision Rule

A failed hard gate moves the initiative to Prepare or Monitor. A strong narrative, senior sponsor or attractive market cannot replace missing authority, evidence, delivery capacity or operating control.

SECTION 01

Readiness Framework

Translate ambition into a shared executive view of value, evidence, capability and risk.



MANAGEMENT ASSESSMENT MODEL

Readiness Dimensions

No.	Decision area	Management evidence	Status
01	Operating requirements	Outcomes, volumes, workflows, exceptions, safety and acceptance	**Confirmed / Developing / Gap**
02	Technology and architecture	Fit, integration, data, identity, infrastructure, scalability and lifecycle	**Confirmed / Developing / Gap**
03	Vendor capability	Experience, team, references, methods, partners, capacity and viability	**Confirmed / Developing / Gap**
04	Risk and assurance	Security, privacy, safety, continuity, data rights, audit and exit	**Confirmed / Developing / Gap**
05	Commercial and delivery	Scope, price, assumptions, change, milestones, warranty, service and remedies	**Confirmed / Developing / Gap**
06	Economics and governance	Lifecycle cost, benefits, sensitivity, contingency, decisions and accountability	**Confirmed / Developing / Gap**

Evidence Standard

Assessments should distinguish verified evidence from management confidence. Where evidence is incomplete, record the gap, owner, completion date and decision that depends on it.

STRATEGIC READINESS DEPENDENCIES

What Readiness Depends On

Dependency	Why it matters	Evidence management should expect
Outcome	The investment decision is explicit	Operating baseline, target, scope, constraints and sponsor
Requirements	Needs are prioritized and testable	Must, should, option, volumes, exceptions and acceptance
Architecture	The solution fits the environment	Systems, data, interfaces, infrastructure, identity and lifecycle
Operational fit	Real workflows can be supported	Normal, peak, degraded, exception, safety and maintenance scenarios
Vendor capability	Delivery and support claims are evidenced	Team, references, methods, capacity, partners and financial viability
Risk	Security, safety and dependency are controlled	Threats, access, data rights, compliance, continuity and exit
Commercial	Terms align incentives and accountability	Scope, price, change, acceptance, warranty, service and remedies
Economics	Lifecycle value is credible	Total cost, ramp, benefits, sensitivity, contingency and residual value

ROALLA Interpretation

Dependencies should be evaluated as a connected system. A weakness in ownership, data, architecture, delivery or governance can change the economics and risk of an otherwise attractive initiative.

SECTION 02

Detailed Readiness Assessment

Convert executive intent into verifiable requirements, accountable evidence and clear remediation actions.



READINESS CHECKLIST

Requirements and Evaluation Readiness

Completion standard Mark Ready only when the supporting evidence has been verified. Assign an evidence owner to every open item.

Gate	Readiness requirement	Ready	Gap	Evidence owner
A01	The investment addresses a verified operating constraint, risk or strategic capability with an accountable sponsor.	☐	☐	_____
A02	Current performance, target outcomes, scope boundaries, assumptions and dependencies are documented.	☐	☐	_____
A03	Requirements are prioritized as mandatory, weighted, optional or informational with clear evidence standards.	☐	☐	_____
A04	Normal, peak, degraded, exception, recovery and future-growth scenarios are included in evaluation.	☐	☐	_____
A05	Architecture requirements cover systems, data, interfaces, identity, infrastructure, telemetry and support.	☐	☐	_____
A06	Safety, cybersecurity, privacy, quality, regulatory and customer obligations are embedded in requirements.	☐	☐	_____
A07	Vendor responses distinguish standard capability, configuration, customization, roadmap and third-party dependency.	☐	☐	_____
A08	Demonstrations use scripted scenarios, client-relevant data and scored acceptance criteria.	☐	☐	_____
A09	Reference checks test comparable scale, complexity, integration, implementation, support and realized outcomes.	☐	☐	_____
A10	Evaluation governance manages conflicts, confidentiality, scoring, clarifications and decision records.	☐	☐	_____

READINESS CHECKLIST CONTINUED

Commercial Delivery and Value Readiness

Completion standard Use Gap to create a dated remediation action and identify the management decision that cannot proceed.

Gate	Readiness requirement	Ready	Gap	Evidence owner
B01	Implementation scope identifies deliverables, assumptions, client responsibilities, exclusions and dependencies.	☐	☐	_____
B02	Milestones and payments align to verified outputs, testing, acceptance and operational readiness.	☐	☐	_____
B03	Change control defines estimate detail, approvals, rates, schedule impact and dispute escalation.	☐	☐	_____
B04	Performance and acceptance tests reflect required capacity, reliability, integration, safety and recovery.	☐	☐	_____
B05	Service terms cover severity, response, resolution, availability, maintenance, spares and escalation.	☐	☐	_____
B06	Data ownership, access, portability, telemetry, model use, confidentiality and deletion are contractually clear.	☐	☐	_____
B07	Security obligations address access, incidents, vulnerabilities, personnel, subcontractors and evidence.	☐	☐	_____
B08	Continuity and exit provisions cover transition assistance, documentation, licences, source and replacement support.	☐	☐	_____
B09	The business case includes acquisition, integration, facility, migration, training, ramp and lifecycle support.	☐	☐	_____
B10	Benefits have baselines, owners, measures, realization actions and management review after implementation.	☐	☐	_____

WEIGHTED EXECUTIVE ASSESSMENT

Readiness Scorecard

Dimension	Weight	Evidence standard	Score	Status
Operating requirements	20%	Outcomes, volumes, workflows, exceptions, safety and acceptance	__/20	Strong / Developing / Gap
Technology and architecture	20%	Fit, integration, data, identity, infrastructure, scalability and lifecycle	__/20	Strong / Developing / Gap
Vendor capability	15%	Experience, team, references, methods, partners, capacity and viability	__/15	Strong / Developing / Gap
Risk and assurance	15%	Security, privacy, safety, continuity, data rights, audit and exit	__/15	Strong / Developing / Gap
Commercial and delivery	15%	Scope, price, assumptions, change, milestones, warranty, service and remedies	__/15	Strong / Developing / Gap
Economics and governance	15%	Lifecycle cost, benefits, sensitivity, contingency, decisions and accountability	__/15	Strong / Developing / Gap
TOTAL	100%	Weighted executive assessment	__/100	Pursue / Prepare / Monitor

Decision Thresholds

Score	Posture	Management decision
75 to 100	**PURSUE**	Proceed to bounded execution subject to no hard-gate failure.
50 to 74	**PREPARE**	Complete a focused readiness plan before major commitment.
0 to 49	**MONITOR**	Resolve foundational gaps before investing further.

A single material hard-gate failure overrides the weighted score. ROALLA also applies judgment where administrative readiness is high but strategic value, adoption or commercial viability is weak.

SECTION 03

Execution and Advisory Roadmap

Build the evidence and evaluation design before inviting vendors, then maintain a clean line from business outcome through requirements, testing, contract and benefit realization.



READINESS OPERATING PLAN

Recommended 90 Day Mobilization

Timing	Focus	Key activities	Management output
Days 1 to 15	Frame	Confirm the decision, outcome, baseline, scope, stakeholders, risks and procurement path.	Approved selection mandate
Days 16 to 35	Specify	Develop prioritized requirements, scenarios, architecture, evidence and scoring.	Decision-ready requirements
Days 36 to 55	Evaluate	Run responses, demonstrations, references, technical, risk and commercial reviews.	Evidence-backed shortlist
Days 56 to 75	Negotiate	Validate scope, plan, team, acceptance, service, data, security, price and terms.	Preferred option and business case
Days 76 to 90	Authorize	Approve contract, governance, funding, benefits, contingency and implementation gates.	Executive award decision

Twelve to Twenty Four Month Path

- Quarter 1: complete selection, contracting, governance and implementation-readiness planning.
- Quarter 2: execute design and build with controlled requirements, change, risk and benefit tracking.
- Quarters 3 and 4: test, accept, transition, stabilize and validate operational and financial outcomes.
- Year 2: manage service, lifecycle, optimization, vendor performance and roadmap decisions using evidence.

ROALLA ADVISORY MODEL

How ROALLA Can Help

ROALLA acts as the client-side technology and transformation advisor throughout industrial selection and investment. ROALLA creates the fact base, requirements, architecture, scoring, business case and governance; coordinates specialist input; challenges vendor claims; and keeps commercial decisions connected to operational outcomes.

ROALLA workstream	Executive value
Investment framing	Define the operating decision, baseline, scope, constraints, outcomes and governance.
Requirements and scenarios	Create prioritized, testable needs across operations, technology, safety, risk and support.
Architecture assurance	Assess systems, interfaces, data, infrastructure, identity, telemetry and lifecycle fit.
Market and vendor evaluation	Manage RFI or RFP, demonstrations, scoring, references and clarifications.
Business case	Model total lifecycle cost, benefits, risk, sensitivity, contingency and funding options.
Commercial advisory	Align scope, milestones, acceptance, change, service, data, security and exit terms.
Implementation governance	Establish decisions, status, risks, dependencies, changes, quality and readiness gates.
Benefit and vendor assurance	Track operational outcomes, service, obligations, lifecycle risk and optimization.

Recommended Initial Mandate

- Four-week industrial technology requirements and vendor-selection mobilization.
- Verified business outcome, current-state, architecture, risk and evaluation fact base.
- Procurement-ready requirements, scripted demonstrations, weighted scorecard and business-case model.
- Executive recommendation and negotiation mandate with implementation decision gates.

CONTROL FRAMEWORK

Key Risks and Controls

Risk	Exposure	ROALLA control
Feature scoring	A long checklist obscures the operating decision.	Weight outcomes, scenarios and evidence over feature count.
Demo theatre	A polished script avoids difficult workflows.	Use controlled client scenarios, data and scoring.
Hidden custom work	Gaps emerge after award as change requests.	Separate standard, configuration, customization and roadmap claims.
Reference mismatch	References differ materially in scale or context.	Use structured, comparable reference validation.
Low bid bias	Implementation and lifecycle costs are understated.	Compare total cost, risk, contingency and sensitivity.
Weak acceptance	Payment occurs before operational value is proven.	Tie milestones to objective tests and readiness evidence.
Data dependency	The vendor controls operational data or portability.	Negotiate ownership, access, formats, export and deletion.
Vendor distress	Capability or support changes after commitment.	Assess viability, partners, continuity and exit protections.

Professional Advice

Qualified legal, privacy, security, tax, accounting, employment and regulatory advisors should validate matters within their professional scope. ROALLA coordinates those inputs with technology strategy, architecture, commercialization, governance and transformation decisions.

OFFICIAL EVIDENCE BASE

Selected Official Sources

These sources support the management principles summarized in this guide. Standards, laws, guidance and organizational obligations continue to evolve and must be confirmed for the specific context before acting.

National Research Council Canada advanced manufacturing initiative. Official source — Active Canadian initiative covering digital manufacturing, sensing, AI, robotics, productivity and sustainable processes.

Canada Advanced Manufacturing Cluster. Official source — Official overview of NGen priorities including automation, machine learning, cybersecurity and SME technology adoption.

Canadian Centre for Cyber Security operational technology guidance. Official source — Canadian guidance for protecting operational technology used in manufacturing and other industrial environments.

NIST Guide to Operational Technology Security. Official source — Risk-based security guidance for industrial control systems and other operational technology.

NIST smart manufacturing systems design and analysis. Official source — Reference architecture, interoperability, analytics and performance-assurance considerations for smart manufacturing.

Ontario Advanced Manufacturing and Innovation Competitiveness Stream. Official source — Ontario support for eligible advanced manufacturing investment in equipment, technology adoption and skills; intake and eligibility must be confirmed.

NRC Industrial Research Assistance Program. Official source — Innovation advisory services, connections and potential financial assistance for eligible Canadian SMEs.

Advisory Disclaimer

This ROALLA guide is a strategic readiness and marketing document. It is not a certification, audit opinion, legal opinion, privacy opinion, security assurance, investment recommendation or guarantee of results. Management must validate current obligations, company facts and professional advice before making material decisions.